Research Gap Matrix

Project: Comparative Analysis of AI-Assisted and Manual Data Labeling Accuracy Across Scientific Datasets **Source analysed:** the attached literature review (section, table and reference numbers below are as printed in it)

Tags: VERIFIED = read in the review text. INFERRED = my reasoning from that text. UNKNOWN = not yet checked against the original publications. **Novelty:** no novelty is claimed for any gap. A separate literature search is required first.

1. Gap matrix

ID	Gap	Passage / reference in the review	Reported in the review (VERIFIED)	Inference (INFERRED)	Experimental question	Evidence that would test it
G1	Matched comparison of AI-assisted vs manual labels against a common reference	Sec. III-A, Table II; workflow studies [9], [10], [11], [13]	Table II mostly reports model-vs-reference Dice. The workflow studies each use a different outcome: time and variability [9], Likert grade [10], Dice and time [11], error rate [13].	The review does not show accuracy of an AI-assisted workflow vs a manual workflow on the same items with the same reference.	On a dataset with an adjudicated reference and several independent annotations, how does agreement with the reference differ between manual labels, model-only labels, and model pre-labels with correction? What does each cost in effort?	Dice/IoU or kappa vs the reference with paired bootstrap confidence intervals; time or query count as cost. Correction step is a human study or a simulation (a weaker proxy).
G2	Size of the anchoring effect, and when it reverses the benefit of AI assistance	Sec. VI-B [19], [20]; Table III (mitigations)	Sec. VI-B describes a paradox of AI accuracy and complacency and the Weight of Advice. Table III lists synthetic-error injection and blind review, uncited.	No effect size is given and the mitigations are untested.	Over a grid of AI accuracy and advice-weighting values, where does assisted accuracy fall below manual accuracy? Does synthetic-error injection move that point?	Simulation sweep with parameters taken from empirical sources (candidates: [19], [20], [63]). Without empirical parameters it only restates assumptions.
G3	Budgeted uncertainty-driven review vs simple baselines	Sec. III-C [11]; Sec. VII-A [22]; Sec. VIII	[11] reports a Dice gain of about 0.01 with large time savings. [22] reports up to a 99% budget reduction, from a conflict-text corpus. The conclusion says the optimal strategy relies heavily on active and confident learning.	The body gives no comparison against random or fixed-fraction review, and one text-domain result supports a general claim.	Under a fixed correction budget, does uncertainty-based routing beat random routing and fixed-fraction review on final label accuracy, across more than one domain?	Budget-vs-accuracy curves over several seeds with confidence intervals, using existing ground truth as the oracle.
G4	Systematic AI-style label errors vs random manual-style errors	Table III; Sec. VII-B; Introduction [1]	Table III says automation bias encodes AI hallucination systematically into ground truth. Sec. VII-B presents Confident Learning as a way to find label errors.	No detection precision or recall is reported, and it is not tested whether detectors built for random noise catch errors correlated with a model's mistakes.	Do Confident-Learning-style detectors find model-correlated label errors as well as random errors at the same rate? How does each noise type affect downstream accuracy?	Inject both noise types into a benchmark; report detection precision and recall and downstream accuracy over several seeds.
G5	Whether reported gains exceed the human noise floor	Sec. II-B; Sec. III-A [8]; Sec. III-C [11]; [27] (listed, never cited)	Quoted gaps are small: HybridMS +0.0103 Dice [11]; GB-SAM +0.028 over U-Net [8].	The review never compares these gaps with the spread between human annotators.	On multi-annotator segmentation data, how large are annotator-to-annotator Dice differences, and do model differences of the reported size fall inside that range?	Pairwise annotator Dice distribution with bootstrap intervals on model differences.

2. Methods, datasets and limitations (basis for the gaps)

Central methods (VERIFIED). Agreement metrics: Cohen's kappa, Fleiss' kappa, Krippendorff's alpha, STAPLE, Dice, IoU (Sec. II). AI-assisted segmentation: zero-shot SAM, MedSAM, GB-SAM, SAM-assisted annotation, AICL, HybridMS (Sec. III). Supervised baseline: nnU-Net. Text and omics: LLM and multi-agent cell-type annotation (Sec. IV-A); Octozi data cleaning (Sec. IV-B). Astronomy: Zoobot, YOLOv5, VLM/VQA (Sec. V-A). Label-noise control: uncertainty-based active learning, ActiveAED/AUM, Confident Learning (Sec. VII).

Datasets (VERIFIED as named; label availability UNKNOWN). Named: BraTS 2023 [7], CRAG [8], Galaxy Zoo [14], Pan-STARRS [16]. Described only: lumbar spine MRI [6], whole-slide pathology pilot [9], CT hemorrhage [10], tuberculosis chest X-ray [11], 34 scRNA-seq datasets [12], ICU consultant data [2], clinical-trial cleaning study [13].

Limitations. The review has no limitations section of its own and states no search protocol, research questions or experiments (VERIFIED). The limitations it reports are of the studied methods: zero-shot SAM domain gap [4]; SAM needs precise box prompts [7]; nnU-Net needs large labeled sets (Table II); subjective manual scRNA-seq annotation [12]; error-prone manual cleaning [13]; crowdsourcing cannot scale [14]; majority voting hides disagreement [1]; no single super expert [2]; fatigue [17], [18]; anchoring and complacency [19], [20]; residual label noise (Sec. VII-B); AI assistance is "not a panacea" (Sec. VIII).

Unresolved (VERIFIED). Out-of-distribution shift is named in the Introduction and in the Sec. VII heading, but no analysis follows; [76]-[79] are never cited. **Unresolved (INFERRED).** No common accuracy measure across domains, no uncertainty on quoted gaps, no measured size of automation bias.

3. Ranking of three gaps

Resources are UNKNOWN. Scores (1-5, higher is better) are provisional (INFERRED), assuming public datasets, modest compute and no human-annotator recruitment.

Rank	Gap	Significance	Resources	Feasibility	Total	Rationale
1	G3	4	5	5	14	Tests the review's own conclusion; runs on small public datasets.
2	G4	4	4	4	12	Tests an uncited claim using noise injection and standard code.
3	G1	5	3	3	11	Matches the title's question; depends on a dataset with per-annotator labels (not checked). Ties G5 on total; ranked higher on significance.

Not ranked: G5 (11) and G2 (9). G5 can be folded into the G1 evaluation as its noise-floor measure. G2 needs verified empirical parameters first.

4. Claims to verify against the original publications

Ref	Check
[9]	29.9 vs 13.6 min and 24% vs 2% variability; number of annotators and slides; the reference title calls it a pilot.
[11]	Whether the 0.0103 Dice gain is tested, and against which baseline.
[13]	Error rates, unit of cleaning error, 6.03x throughput.
[10]	What graders rated, whether blinded; the review's explanation that AI uncovers boundary details hidden by fatigue is untested in its text.
[22]	Whether the 99% figure is from this paper and generalizes beyond conflict text corpora.
[2]	Kappa 0.383 and 0.255; eleven consultants.
[6], [7], [8]	Dice values; that nnU-Net 0.99 is lumbar only.
[19], [20], [63]	Whether they report anchoring effect sizes usable for G2.
ActiveAED / AUM	Cited as [23] in the text; the list has it at [71]. Confirm source and claim.
Confident Learning	Cited as [21], a retinal-image application. Check whether the originating paper is in the list.
Uncited items	Oral-lesion kappa 0.90; "substantial label noise in curated benchmarks"; Sec. V-A VLM/VQA statement.

All numbers the review attributes to its sources are UNKNOWN until confirmed in the originals.